

PPC to SMA Inverter — Modbus Sequence Diagram

Scope: OB30 cycle (100 ms): Read Modbus → Process PPC logic → Write Modbus, per inverter

Unit ID: 3 | Protocol: Modbus TCP | Platform: SIMATIC S7 (TIA Portal)

1. OB30 Cycle Flowchart

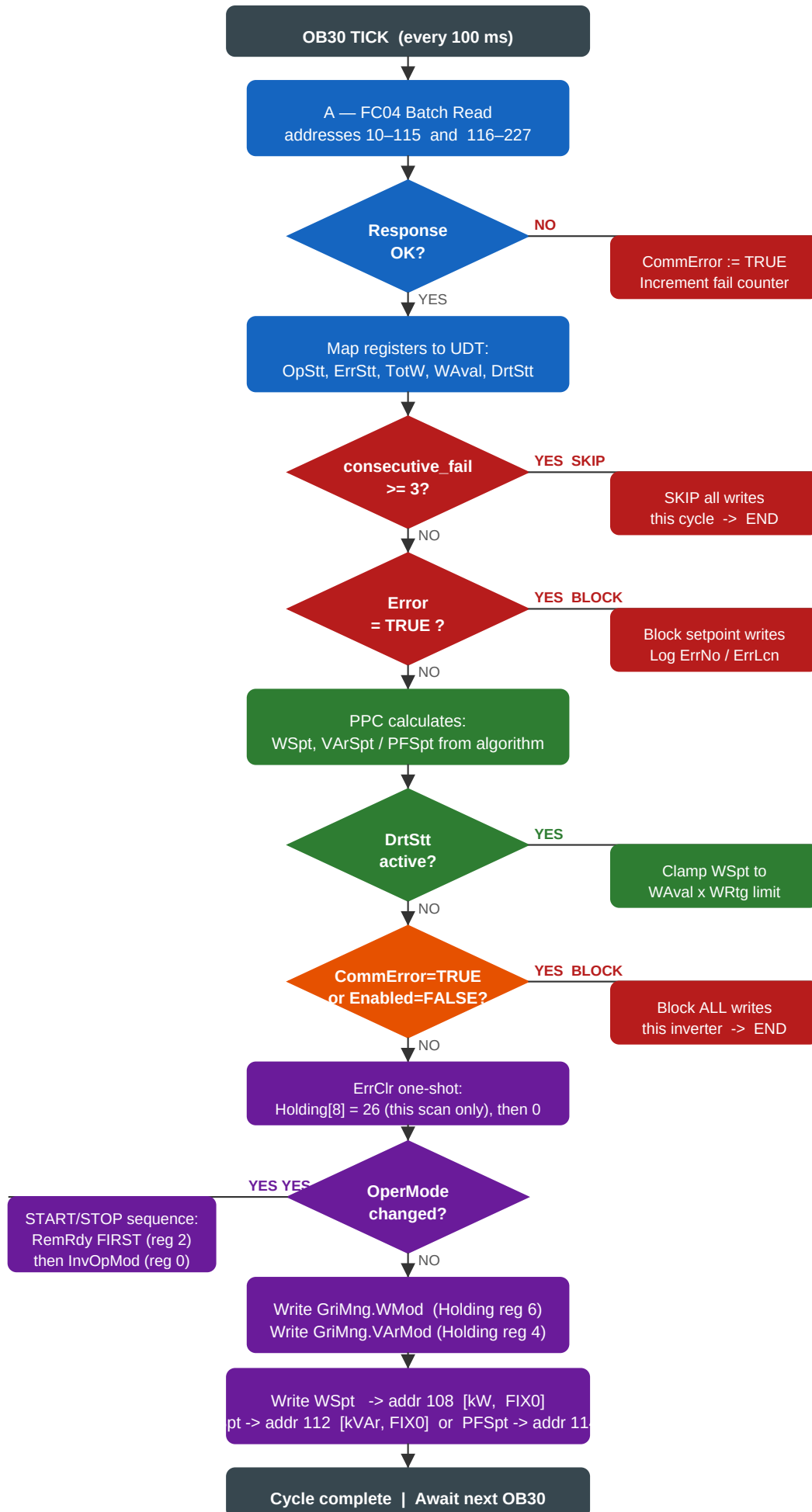
Blue = Read phase | Green = Process | Orange = Interlock | Purple = Write | Red = Error/Block

A READ

B PROCESS

C INTERLOCK

D WRITE



2. Startup Sequence

Execute once when plant enable is received and CommError = FALSE.

Step	Trigger / Condition	Read-back to check	Write register	Value	Interlock / Fail-safe
1	Plant enable + CommError=FALSE	ErrStt=307 (OK); OpStt not Error; KeySw=On	GriMng.WMod (6) GriMng.VArMod (4)	1079 WCtlCom 1072 VArCtlCom	Block if ErrStt=Error
2	Enable ramps — write once at startup	—	WGraMod (12) VArGraMod (16)	308 Enable 308 Enable	If rejected, PPC must ramp externally
3	Soft-start setpoints (avoid large step)	WAval / VArAval available	WSpt (108) VArSpt (112)	0 kW 0 kVAr	Alarm if setpoint rejected
4	Command READY	No blocking PwrOffReas	RemRdy (2)	308 Ready	Must be READY before OPERATION
5	Command OPERATION	OpStt transitions toward GridFeed (3526)	InvOpMod (0)	308 Operation	Alarm + retry if OpStt=Error after 60 s
6	Closed-loop PPC dispatch	OpStt = 3526 GridFeed	WSpt (108) VArSpt/PFSpt (112/114)	Per PPC schedule	Keep control modes stable; monitor DrtStt

3. Stop Sequence

TRIGGER: OperMode = 303 (PPC command or operator)

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Step 1  Ramp WSpt to 0 over P_RampDown period
Step 2  Write addr [108] = 0                (WSpt = 0 kW)
Step 3  Write Holding[2] = 303              (RemRdy = Standby) <- FIRST
Step 4  Write Holding[0] = 303              (InvOpMod = Stop)
Step 5  READ OpStt -> wait for 381 (Stop) or 3528 (Standby)
```

4. Fault Handling Sequence

DETECT: ErrStt = 1392 (Error) OR OpStt = 1392

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Step 1  Set Error := TRUE in UDT
Step 2  Read ErrNo (reg 96) -> log fault code to SCADA
Step 3  Read ErrLcn (reg 92) -> log fault location
Step 4  Read PwrOffReas (reg 178) -> display disconnect reason
Step 5  Block WSpt / VArSpt writes while Error = TRUE
Step 6  OPERATOR: investigate and clear fault cause
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Step 7  ACKNOWLEDGE (one-shot – exactly ONE OB30 scan):
        Set UDT.ErrClr := 26
        Comms block writes Holding[8] = 26
        Next scan: Holding[8] = 0 (rising edge required)
        Safety-critical ProErr: also write Holding[20] = 973
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Step 8  Monitor ErrStt -> if = 307 (OK): Error := FALSE
Step 9  Restart from startup sequence Step 1
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5. Communication Error / Timeout

DETECT: FC04 no response within Modbus timeout

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consecutive_fail := consecutive_fail + 1
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IF consecutive_fail >= 3 THEN
  CommError      := TRUE
  AnyFault       := TRUE          (plant level)
  FaultMask.bit[i] := TRUE
  Block ALL writes to this inverter
  Remove inverter from PPC dispatch (reduce N_online)
  Hold last valid UDT values        (do NOT zero setpoints)
END_IF
```

```
RECOVERY: FC04 request succeeds
  CommError      := FALSE
  consecutive_fail := 0
```

6. Interlock Summary

Condition	Effect on Writes	Source
CommError = TRUE	Block ALL writes to this inverter for this scan	Internal UDT flag
Enabled = FALSE	Block ALL writes to this inverter	DB39.Skid[i].Enabled
ErrStt = 1392 (Error)	Block setpoint writes: WSpt, VARspt, PFSpt	Input reg 94
OperMode = 308 (Start)	Write RemRdy=308 (reg 2) BEFORE InvOpMod=308 (reg 0) — same FC16 frame preferred	Holding regs 2 + 0
OperMode = 303 (Stop)	Write RemRdy=303 (reg 2) BEFORE InvOpMod=303 (reg 0)	Holding regs 2 + 0
ErrClr != 0	Write Holding[8]=26 THIS scan only; write 0 every subsequent scan	Holding reg 8
DrtStt active (not 973)	Clamp WSpt to WAval x Wrtg (inverter availability constraint)	Input reg 176
PwrOffReas = 21626	Increase WSpt/VARspt above inverter minimum threshold	Input reg 178
OpStt = 3527 (FRT)	Do NOT issue Stop command during fault ride-through	Input reg 98
OpStt = 1787 (Init)	Hold all writes; wait for stable state before commanding	Input reg 98

7. Register Quick Reference

Dir	Channel	Addr	FC	Fmt	Unit	Purpose
WRITE FC16	RemRdy	2	FC16	ENUM	—	Ready(308)/Standby(303) — write BEFORE InvOpMod
WRITE FC16	InvOpMod	0	FC16	ENUM	—	Operation(308) / Stop(303)
WRITE FC16	GriMng.VArMod	4	FC16	ENUM	—	VArCtlCom(1072) / PFCtlCom(1075) / Off(303)
WRITE FC16	GriMng.WMod	6	FC16	ENUM	—	WCtlCom(1079) / Off(303)
WRITE FC16	ErrClr	8	FC16	ENUM	—	Fault acknowledge — one-shot: write 26 then 0
WRITE FC16 *	WSpt	108	FC16	FIX0	kW	Active power setpoint
WRITE FC16 *	VARspt	112	FC16	FIX0	kVAr	Reactive power setpoint
WRITE FC16 *	PFSpt	114	FC16	FIX4	—	Power factor x10000 (9500=0.95, 10000=1.00)
READ FC04	InvMs.TotW	28	FC04	FIX0	kW	Measured AC active power
READ FC04	InvMs.TotVAr	30	FC04	FIX0	kVAr	Measured AC reactive power
READ FC04	ErrStt	94	FC04	ENUM	—	307=OK, 1392=Error
READ FC04	ErrNo	96	FC04	FIX0	—	Fault code — log when Error=TRUE
READ FC04	OpStt	98	FC04	ENUM	—	3526=GridFeed, 3527=FRT, 3528=Standby, 1392=Error
READ FC04	WSpt (readback)	108	FC04	FIX0	kW	Active setpoint currently in effect
READ FC04	VARspt (readback)	112	FC04	FIX0	kVAr	Reactive setpoint currently in effect
READ FC04	WAval	172	FC04	FIX4	pu	Available active power — divide by 10000

Dir	Channel	Addr	FC	Fmt	Unit	Purpose
READ FC04	VARAval	174	FC04	FIX4	pu	Available reactive power
READ FC04	DrtStt	176	FC04	ENUM	—	973=no derating, 21586=FRT, 21601=WCtlVol
READ FC04	PwrOffReas	178	FC04	ENUM	—	21626=Low Power SetPoint
READ FC04	WRtg	184	FC04	FIX0	kW	Rated power — read once at startup

* WSpt/VARSp/PFSpt: Input Register addresses (FC04 readback). SMA Sunny Central also accepts FC16 writes to these same addresses when WCtlCom / VARCtlCom / PFCtlCom mode is active.